

1.4 Classwork Solutions: Measurement quiz review

Notation: = joins values that are exactly equal; $\approx$ introduces a rounded value. Students should write it the same way — once you round the calculator display, the equals sign is no longer true.

1. Significant digits.

(a) **3**

(c) **2**

(e) **4**

(b) **2**

(d) **3**

(f) **2**

(b) The zero in 3.0 is after the decimal point, so it was measured. (c) The leading zeros in 0.045 only locate the decimal point. (d) A zero between nonzero digits always counts. (e) Both zeros count: the middle one is sandwiched, the trailing one is after the decimal point. (f) In scientific notation, every digit of the coefficient counts and nothing else does.

2. Round to three significant figures.

(a) $\sqrt{10} = 3.1622776 \dots \approx \mathbf{3.16}$

(b) $\frac{1}{7} = 0.1428571 \dots \approx \mathbf{0.143}$

The leading zero is not significant, so the three figures are 1, 4, 2 — and the next digit, 8, rounds the 2 up. A common error is 0.14, which is only two figures.

3. Scientific notation.

(a) $149,600,000 \text{ km} = 1.496 \times 10^8 \text{ km} \approx \mathbf{1.50 \times 10^8 \text{ km}}$

Write 1.50, not 1.5 — the trailing zero is the third significant figure.

(b) $0.000102 \text{ m} = \mathbf{1.02 \times 10^{-4} \text{ m}}$ (already exactly three figures — nothing to round)

Count the places the decimal point moves: four, to the right, so the exponent is negative. That is 0.1 mm, a tenth of a millimeter.

4. Ordinary decimal numbers.

(a) $8.849 \times 10^3 \text{ m} = \mathbf{8849 \text{ m}}$

(b) $1.0 \times 10^{-3} \text{ kg} = \mathbf{0.0010 \text{ kg}}$

Keep the trailing zero: 1.0×10^{-3} has two significant figures, and so must the decimal. Writing 0.001 kg is the right size but drops a figure. This is also exactly 1 gram.

5. Reading the ruler. The bar ends between the 2nd and 3rd millimeter marks past 4 cm, closer to the 3rd.

(a) **4.28 cm** — the 4.2 is certain, and the final 8 is the estimated digit.

Accept 4.27 to 4.29. Do not accept 4.2 or 4.3 (no estimated digit).

$$(b) \quad 4.28 \text{ cm} \times \frac{10 \text{ mm}}{1 \text{ cm}} = \mathbf{42.8 \text{ mm}} \quad 4.28 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} = \mathbf{0.0428 \text{ m}}$$

All three have three significant figures — changing units never changes the precision.

6. Convert.

$$(a) \quad 56.7 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 185.976 \text{ ft} \approx \mathbf{186 \text{ ft}}$$

Check the direction: a foot is shorter than a meter, so the number of feet must be larger than the number of meters.

$$(b) \quad 11.0 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} = 27.94 \text{ cm} \approx \mathbf{27.9 \text{ cm}}$$

$$7. \quad v = \frac{d}{t} = \frac{60.0 \text{ m}}{45.0 \text{ s}} = 1.3333 \dots \approx \mathbf{1.33 \text{ m/s}}$$

A normal walking pace is about 1.4 m/s, so this is a relaxed stroll.

$$8. \quad 119 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{119,000}{3600} \frac{\text{m}}{\text{s}} = 33.055 \dots \approx \mathbf{33.1 \text{ m/s}}$$

Both the kilometers and the hours cancel, leaving meters over seconds. A quick check: dividing km/h by 3.6 gives m/s.